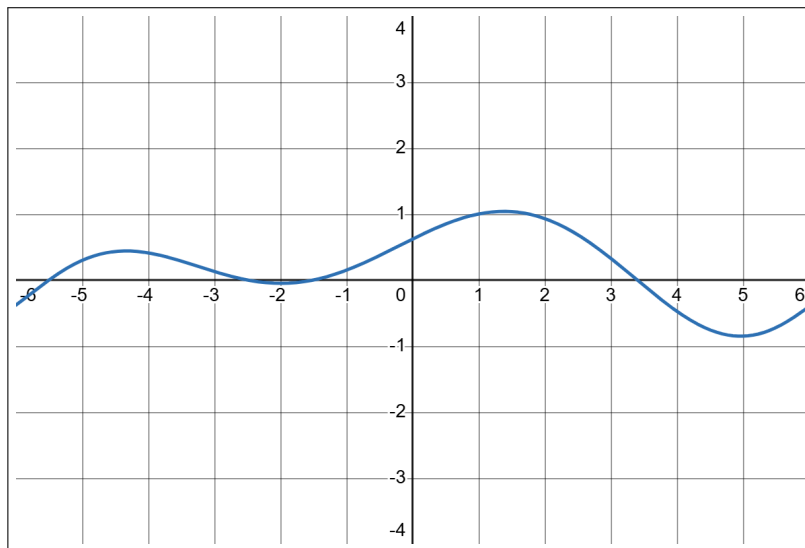


A4 past final worksheet

1. (Fall 2024, #4) Consider the function f whose graph is shown below.



- (a) One of the following is the exact value of $f'(1)$. Pick which it is. Note: The peak of the graph occurs a little to the right of $x = 1$, not at $x = 1$.
- i. 100
 - ii. $\frac{1}{5}$
 - iii. $-\frac{1}{5}$
 - iv. -300
- (b) Use only your answer to part (a) and the fact that $f(1) = 1$ and no other information to estimate $f(0.9)$.
- (c) Determine whether your approximation in (b) is an over or underestimate of the actual value of $f(0.9)$ and justify your answer. (Hint: use the given graph).

2. (Spring 2024, #4) Consider the equation $\cos(xy) = x^2 + y$ and the curve it defines.

(a) Find an equation for the tangent line to the curve at the point $(1, 0)$.

(b) Use your answer (and the idea of linearization) to find an approximation for a y so that $(1.1, y)$ is on the curve.

3. (Fall 2023, #4) Given the relation

$$e^{xy} + y \sin x + y^2 - 10 = 0$$

- (a) Find $\frac{dy}{dx}$ at $(0, 3)$.

- (b) Find the equation of the tangent to its graph at $(0, 3)$.

4. (Spring 2023, #3) Let $f(x) = (2x + 2)^{\frac{2}{3}}$.

(a) Find the linearization of $f(x)$ at $x = 3$.

(b) Use your linearization from part (a) to approximate $(8.04)^{\frac{2}{3}}$.

5. (Spring 2022, #5) Consider the curve $e^{xy} + 3x^2 + y^2 = 65$.

(a) Find the equation of the tangent line at the point $(0, 8)$.

(b) There are two points on the curve with x -coordinate 0.1. Use part (a) to estimate one possible y -coordinate of the point $(0.1, y)$.